System Validation Report

Outcome: **PASS** — 122 passed, 0 failed.

Run: 2026-11-12T16:49:13.470921+00:00 to 2026-11-12T16:49:15.211579+00:00

Source SHA: 6b5464298c949806ebba0b7bd1d903c962b5255d (finish 6b5464298c949806ebba0b7bd1d903c962b5255d)

Binary: /home/deeptanshukumar/Downloads/SE-Railway Ticket Reservation System/build-submission/rtrs_server

Binary SHA-256: 8437a1bb68a34e759244665e3c4deb2e565bc843e6ba2d53e46cd6e2a724f451

Travel date: 2026-11-15 (system local date +3 days).

Environment


```
{
"platform": "Linux-7.2.14-200.fc44.x86_64-x86_64-with-glibc2.43",
"python": "3.13.13 | packaged by Anaconda, Inc. | (main, Apr 14 2026, 06:19:41) [GCC 14.3.0]",
"openssl": "OpenSSL 3.6.3 9 Jun 2026",
"cpuCount": 18,
"timezone": [
"IST",
"IST"
],
"pythonSqliteVersion": "3.51.2"
}
```


The Python SQLite version above describes the runner’s sqlite3 module, not the server. The application bundles SQLite 3.46.1; this is not a runtime server-version measurement.

Method and scope

isolation: Fresh TemporaryDirectory under /tmp, dedicated SQLite DB; inherited RTRS_* settings removed; ephemeral ports; no production DB

transport: Actual server subprocess and localhost HTTP/HTTPS sockets; no mocks

load: 20 search + 20 two-passenger bookings, separately, four concurrent clients, fresh connection per request, nearest-rank p90 <3 seconds each; connection through full response latency

recovery: Controlled restart and abrupt process kill/relaunch after committed writes, same database plus read-only SQLite integrity check; not power-loss or interrupted-transaction durability

credentials: Random runtime password; request bodies and auth headers never logged by runner; response secrets redacted

Measured performance

Operation	Requests	Clients	p90 (s)	Max (s)	Threshold
search	20	4	0.002256	0.002906	<3 s
booking	20	4	0.006130	0.007966	<3 s

Detailed expected/actual bodies, raw latency samples and per-step timestamps are preserved in raw JSON evidence. Check IDs below identify the corresponding 1-based entries in its steps array.

Failures and limitations

No failed assertions observed in executed checks.

Small local load only; no sustained stress or internet latency conclusions

TLS uses a temporary self-signed trust anchor; does not validate production certificates

Admin audit API records booking events; train changes verified via read-back and restart

No UI, screenshot, coverage or unit-test results claimed by this independent runner

Check results

Check ID	Name	Status
SV-001	Start HTTP server	PASS
SV-002	Register owner@example.test	PASS
SV-003	Registration cannot assign admin: owner@example.test	PASS
SV-004	Register other@example.test	PASS
SV-005	Registration cannot assign admin: other@example.test	PASS
SV-006	Register admin@example.test	PASS
SV-007	Registration cannot assign admin: admin@example.test	PASS
SV-008	Login owner@example.test	PASS
SV-009	Login other@example.test	PASS
SV-010	Login admin@example.test	PASS
SV-011	Anonymous profile denied	PASS
SV-012	Profile update	PASS
SV-013	Profile read	PASS
SV-014	Profile update and role protection	PASS
SV-015	Future Pune to Mumbai search	PASS
SV-016	Seeded route results	PASS
SV-017	Schedule	PASS
SV-018	Ordered route stops	PASS
SV-019	Seat inventory	PASS
SV-020	Class fares	PASS
SV-021	Fare consistency	PASS
SV-022	Booking preview	PASS
SV-023	Preview fare arithmetic	PASS
SV-024	Seat inventory	PASS
SV-025	Seat inventory	PASS
SV-026	Preview does not reserve	PASS
SV-027	Create booking	PASS
SV-028	Confirmed ticket and assigned unique seats	PASS
SV-029	Seat inventory	PASS
SV-030	Seat inventory	PASS
SV-031	Seat decrement	PASS
SV-032	Owner ticket	PASS
SV-033	Ticket matches confirmed booking	PASS
SV-034	Owner history	PASS
SV-035	History contains booking	PASS
SV-036	Cross-user denied GET	PASS
SV-037	Cross-user denied GET/cancellation	PASS
SV-038	Cross-user denied POST/cancel	PASS
SV-039	Cross-user denied PUT/cancel	PASS
SV-040	Other user history	PASS

Check ID	Name	Status
SV-041	History isolation	PASS
SV-042	Refund quote	PASS
SV-043	Default cancellation policy	PASS
SV-044	Cancel booking	PASS
SV-045	Refund recorded and cancellation timestamp	PASS
SV-046	Repeat cancellation rejected	PASS
SV-047	Seat inventory	PASS
SV-048	Seats restored exactly once	PASS
SV-049	Passenger admin denial /api/admin/me	PASS
SV-050	Anonymous admin denial /api/admin/me	PASS
SV-051	Passenger admin denial /api/admin/trains	PASS
SV-052	Anonymous admin denial /api/admin/trains	PASS
SV-053	Passenger admin denial /api/admin/audit	PASS
SV-054	Anonymous admin denial /api/admin/audit	PASS
SV-055	Passenger admin write denied	PASS
SV-056	Stopped-server admin promotion CLI	PASS
SV-057	Start HTTP server	PASS
SV-058	Promotion revokes old session	PASS
SV-059	Login admin@example.test	PASS
SV-060	Admin role	PASS
SV-061	Promoted role	PASS
SV-062	Admin create train	PASS
SV-063	Admin update train	PASS
SV-064	Admin update reflected	PASS
SV-065	Created train public visibility	PASS
SV-066	Managed train searchable	PASS
SV-067	Admin deactivate train	PASS
SV-068	Train inactive	PASS
SV-069	Inactive train excluded from search	PASS
SV-070	No inactive results	PASS
SV-071	Inactive schedule unavailable	PASS
SV-072	Admin audit	PASS
SV-073	Booking audit correlates actor, booking, refund	PASS
SV-074	Malformed JSON	PASS
SV-075	Malformed JSON safe error	PASS
SV-076	Invalid booking passengers	PASS
SV-077	Invalid booking passengers safe error	PASS
SV-078	Invalid date	PASS
SV-079	Invalid date safe error	PASS
SV-080	Unknown endpoint	PASS
SV-081	Unknown endpoint safe error	PASS
SV-082	Invalid audit limit	PASS
SV-083	Invalid audit limit safe error	PASS
SV-084	Anonymous booking	PASS

Check ID	Name	Status
SV-085	Anonymous booking safe error	PASS
SV-086	Unknown account login	PASS
SV-087	Wrong password login	PASS
SV-088	Login does not enumerate accounts	PASS
SV-089	search small-load p90	PASS
SV-090	booking small-load p90	PASS
SV-091	Concurrent booking unique PNRs and seats	PASS
SV-092	Pre-restart history	PASS
SV-093	Start HTTP server	PASS
SV-094	Login owner@example.test	PASS
SV-095	Persisted profile	PASS
SV-096	Persisted bookings	PASS
SV-097	Restart persistence exact history/profile	PASS
SV-098	Seat inventory	PASS
SV-099	Seat inventory	PASS
SV-100	Restart inventory persists	PASS
SV-101	Login admin@example.test	PASS
SV-102	Persisted admin train	PASS
SV-103	Persisted audit	PASS
SV-104	Admin state and audit survive restart	PASS
SV-105	Abrupt process stop	PASS
SV-106	Start HTTP server	PASS
SV-107	Login owner@example.test	PASS
SV-108	Crash-recovered history	PASS
SV-109	Committed bookings survive abrupt stop	PASS
SV-110	Seat inventory	PASS
SV-111	Seat inventory	PASS
SV-112	Crash-recovered inventory	PASS
SV-113	Recovered SQLite integrity	PASS
SV-114	Logout	PASS
SV-115	Logged-out session denied	PASS
SV-116	Generate temporary TLS certificate	PASS
SV-117	Start HTTPS server	PASS
SV-118	Real verified TLS handshake	PASS
SV-119	HTTPS health and security headers	PASS
SV-120	Login owner@example.test	PASS
SV-121	Authenticated profile over real TLS	PASS
SV-122	Server logs contain no generated credentials	PASS